Yuqing Zhou

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Research Interests

Self-evolving AI, LLM Reasoning, LLM Post-training, Trustworthy AI

Education

George Mason University — **Aug. 2023 – May 2028 (Expected)**
Ph.D. in Computer Science — *Fairfax, VA, USA*

University of Michigan — **Aug. 2019 – Apr. 2021**
M.S. in Electrical and Computer Engineering, Computer Vision — *Ann Arbor, MI, USA*

Southeast University — **Aug. 2015 – Jun. 2019**
B.Eng. in Electronic Science and Technology — *Nanjing, Jiangsu, China*

Work Experience

Applied Scientist Intern, Amazon Web Services (AWS) — **May 2026 – Aug. 2026**
Efficient LLM Reasoning — *Seattle, WA, USA*

- Developed a step-aware credit assignment method for RL-based post-training that redistributes trajectory-level learning signals according to the contribution of individual reasoning steps, preserving useful computation while suppressing redundant and dead-end reasoning.
- On Qwen2.5-Math-7B, outperformed trajectory-level length regularization, trace-compression, and fine-grained credit-assignment baselines, reducing reasoning tokens by **24% on average** while maintaining or improving accuracy.

Applied Scientist Intern, Amazon Web Services (AWS) — **May 2025 – Aug. 2025**
Agentic Conversational AI — *Seattle, WA, USA*

- Developed workflow-based control mechanisms for tool-using conversational agents, representing procedures as structured memory with prerequisite-aware action specifications.
- Built an offline-to-online learning pipeline that induces workflows from historical agent trajectories and uses runtime progress alignment to guide agent execution.
- Evaluated the approach on interactive agent benchmarks including ToolSandbox and τ^2-bench, consistently improving agent performance and execution reliability over baseline agents.

Software Engineer, Shanghai Huawei Technologies Co., Ltd, — **Aug. 2021 – May 2023**
Shanghai, China

Publications

[1] **Yuqing Zhou**, Zhuoer Wang, Jie Yuan, Hong Wang, Samson Koelle, Ziwei Zhu, and Wei Niu. "WISE-Flow: Workflow-Induced Structured Experience for Self-Evolving Conversational Service Agents". In: *Findings of AACL* (2026).

[2] Bowen Wei, Nan Wang, **Yuqing Zhou**, Jinhao Pan, and Ziwei Zhu. "Confidence-Orchestrated Self-Evolution against Uncertain LLM Feedback". In: *EMNLP* (2026).

[3] **Yuqing Zhou** and Ziwei Zhu. "Fighting Spurious Correlations in Text Classification via a Causal Learning Perspective". In: *NAACL* (2025).

[4] **Yuqing Zhou**, Ruixiang Tang, Ziyu Yao, and Ziwei Zhu. "Navigating the Shortcut Maze: A Comprehensive Analysis of Shortcut Learning in Text Classification by Language Models". In: *Findings of EMNLP* (2024).

[5] **Yuqing Zhou**, Tianshu Feng, Mingrui Liu, and Ziwei Zhu. "A Generalized Propensity Learning Framework for Unbiased Post-Click Conversion Rate Estimation". In: *CIKM* (2023).

Selected Research Projects

Causally Calibrated Robust Classifier (CCR) **Mar. 2024 – Oct. 2024**

- Developed a group-label-free framework for improving worst-group robustness under distribution shift by reducing model reliance on spurious correlations.
- Designed a causal feature selection mechanism together with inverse propensity weighting to identify and downweight unstable predictive correlations without requiring group annotations.
- Achieved **70.7% worst-group accuracy** on CivilComments, outperforming competing robust-learning methods that require explicit group labels.

Shortcut Learning Analysis in Language Models **Sep. 2023 – Jun. 2024**

- Developed a systematic taxonomy of shortcut learning in language models, covering occurrence-, style-, and concept-based shortcuts.
- Constructed controlled benchmarks for these failure modes, including LLM-generated datasets with synthetic shortcut correlations.
- Evaluated BERT and Llama-family models, including Llama2-7B/13B and Llama3-8B, together with state-of-the-art robust learning methods, showing that increased model scale alone does not consistently improve shortcut robustness.

Generalized Propensity Learning for Post-Click Conversion Rate Prediction **May 2023 – Jun. 2023**

- Developed a generalized propensity-learning (GPL) framework to minimize bias and variance in post-click conversion prediction (CVR) for recommender systems, enhancing the performance and robustness of existing methods like IPS and DR-based estimators.
- Improved CVR prediction accuracy by **7% in DCG@2** and **6% in Recall@2** on the Yahoo dataset.

Technical Skills

Programming: Python, C/C++, Julia, MATLAB

ML Frameworks: PyTorch, Hugging Face Transformers
LLM Training & Inference: verl, TRL, vLLM

Teaching Experience

Graduate Teaching Assistant, George Mason University

CS 657: Mining Massive Datasets — Fall 2026
CS 584: Theory and Applications of Data Mining — Fall 2026
CS 455: Computer Communications and Networking — Fall 2024, Spring 2025

Academic Service

Reviewer: ICLR (2025–2027), KDD (2026–2027), PAKDD (2025), AISTATS (2025), IEEE TKDE, ACM TOIS, ACM TIST, ACM TORS